Symptom:

P0457-LOOSE FUEL FILL CAP

When Monitored and Set Condition:

P0457-LOOSE FUEL FILL CAP

When Monitored: Engine Running. Cold start test. Fuel Level greater than 12%. Ambient Temperature between 4°C and 32°C (39°F and 89°F) Close Loop fuel system. Test runs when small leak test is maturing.

Set Condition: The PCM activates the EVAP Purge Solenoid to pull the EVAP system into a vacuum to close the NVLD switch. Once the NVLD switch is closed, the PCM turns the EVAP Purge Solenoid off to seal the EVAP system. If the NVLD switch reopens before the calibrated amount of time after a fuel tank fill, an error is detected. Two Trip Fault. Three good trips to turn off the MIL.

POSSIBLE CAUSES
GOOD TRIP EQUAL TO ZERO
LOOSE OR MISSING FUEL FILL CAP
INTERMITTENT CONDITION
NVLD SERVICE TEST
EVAPORATIVE EMISSION LEAK DETECTION
EVAP PURGE SOLENOID OPERATION
NVLD SWITCH OPERATION

TEST	ACTION	APPLICABILITY
1	**NOTE: After the PCM has determined the leak test inconclusive and sees an increase in fuel level, the PCM will request the GAS CAP indicator on to inform the customer that the gas fill cap is loose or off.** Ignition on, engine not running. With the DRBIII®, read DTCs and record the related Freeze Frame data. Is the Good Trip Counter displayed and equal to zero? Yes → Go To 2 No → Refer to the INTERMITTENT CONDITION symptom in the Driveability category. Perform POWERTRAIN VERIFICATION TEST VER - 6.	All